Final Layout Submission (17/10/2026)

Initial Report

Course: UBCx: Silicon Photonics Design, Fabrication and Data Analysis

Professor: Lukas Chrostowski

Layout Topic: A scalable WDM-based optical transmission system for signal transport based on ring resonators

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According to the feedback received for the draft layout, two MZIs are also included in the layout even though they weren't the main design, they are just added to quarantee any possible achievable results from my final layout. These MZIs own delta_length of 60um and 80um between their routings, which with group index of about 4.204 and wavelength of 1550nm, about 9.09nm and 6.82nm FSR are expected to be acieved, the possible results are calculated using the following equation.

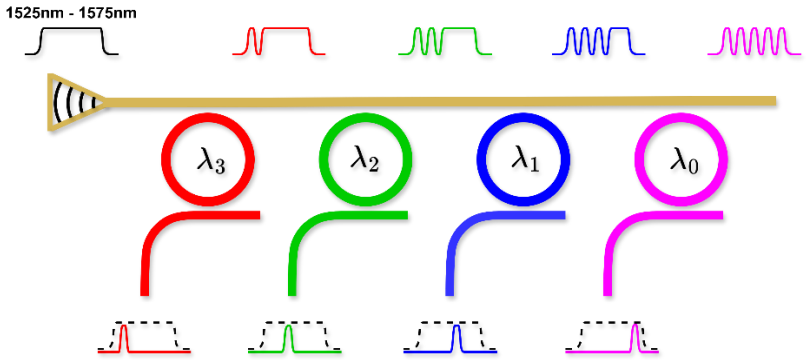
$$FSR = \frac{\lambda^2}{\Delta L \times n_g}$$

However Ring Resonators are the fundamental element of this layout. They serve as the wavelength-selective filters [1], [2], enabling the extraction of distinct optical channels from the laser spectrum. To ensure effective channel filtering, the Free Spectral Range (FSR) and Full Width at Half Maximum (FWHM) of these rings need to be optimized. A larger FSR helps to prevent resonance overlap within the laser spectrum, but this usually demands smaller ring radii, which can degrade the quality factor (Q) due to increased bending losses. A lower Q leads to broader resonance peaks and less selective filtering.

To overcome this issue, trade-off between FSR and FWHM needs to be satisfied for the ring resonators, with proper design, RRs will provide highly selective, compact, and passive spectral slicing that is essential for multi-channel operation. The 4-bit ring resonator bank which is employed to test the realisibility of the modulator is also represented in the next page.

The drop port of each resonator connects to a combiner in order to recombine modulated channels, the composite signal, which includes all active WDM channels, is transmitted through an optical fiber.

Resonator Banks implemented in this layout



2 ring resonator bank:

ring radii 15um and 14.94um, coupling length 2.5um/5um

3 ring resonator bank:

ring radii 15um, 14.96um and 14.92, coupling length 2.5um/5um

4 ring resonator bank:

ring radii 15um, 14.97um, 14.94 and 14.91, coupling length 2.5um/5um

Largest FSR for these resonators are about 5nm and the difference in the radii are chosen in a way which they are located in fairly equal distance from each other in a 5nm spectrum.

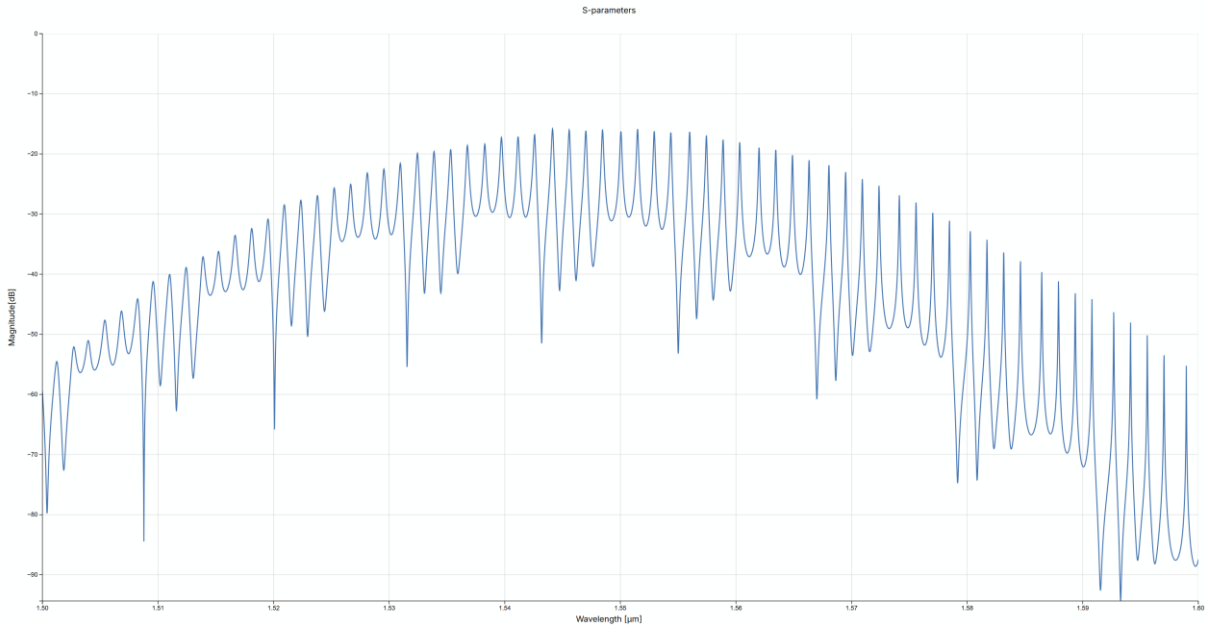
$$FSR = \frac{\lambda^2}{L \times n_g}$$

L in this formul is the ring perimeter,

$$L = 2 \times L_{coupling} + 2\pi R$$

R is radius, for our about 5um coupling length, the L will be approximately 104.2um, and FSR will be equal to 5.5nm, the important issue here is that as much as wavelength slices are fit inside this amount of FSR with a fair gap between them and just to make sure that their bandwidths are sharp enough to avoid overlaps between the channels, that's the reason why two numbers for coupling length are chosen, just to make sure that the energy will have this oppurtunity to stay more inside the ring before it couples back in the main bus because of high coupling length.

Estimated result for the 4 RR bank is represented below, the result is achieved using Luceda Canvas:



References:

1] W. Bogaerts, P. DeHeyn, “Silicon microring resonators”, Laser Photonics Rev. 6, No.1, 47–73 (2012)/DOI 10.1002/lpor.201100017

[2] Geuzebroek, D.H., Driessen, A. (2006). “Ring-Resonator-Based Wave length Filters”, Springer Series in Optical Sciences, vol 123. Springer, Berlin, Heidelberg. doi.org/10.1007/3-540-31770-8-9

[3] R. Boeck, N. A. F. Jaeger, ”Series-coupled silicon racetrack resonators and the Vernier effect: theory and measurement,” Opt. Express 18, 25151 25157 (2010)

[4] Peng, Z.; Arakawa, T. “Tunable Vernier Series-Coupled Microring Res onator Filters Based on InGaAs/InAlAs Multiple Quantum-Well Waveg uide”. Photonics 2023